OEA r0.3 — 3.3 V / 5 V HOST TO EXTERNAL STEP/DIR/ENABLE

PROTOTYPE CANDIDATE — NOT A MANUFACTURING RELEASE. Exact host/driver profiles require qualification.

External power required in BOTH modes: 5 V input uses EXT only, 12 V / 24 V input may use regulated 5V or EXT.

Power off before selecting 5V/EXT. Shared ground, no isolation. VM omitted: HOST VIO NC. Commands use direct MOSFET sinks.

BUCK_VIN

GND

V_IND_IN

C50
100nF 50V X7R
V_IND_3V3 1
2 GND

C51
100nF 50V X7R
IND_ADC 1
2 GND

D50
5V red LED
GND 1
K
A 2 IND_LED5_A

D51
12V red LED
GND 1
K
A 2 IND_LED12_A

D52
24V red LED
GND 1
K
A 2 IND_LED24_A

R50
1k 1%
IND_LED_5V 1
2 IND_LED5_A

R51
1k 1%
IND_LED_12V 1
2 IND_LED12_A

R52
1k 1%
IND_LED_24V 1
2 IND_LED24_A

R53
100k 0.1% 25ppm/C
COM_PLUS 1
2 IND_ADC

R54
3.9k 0.1% 25ppm/C
IND_ADC 1
2 GND

TP50
UPDI
IND_UPDI 1
UPDI

TP51
3V3 SENSE ONLY
V_IND_3V3 1
3V3

TP52
GND
GND 1
GND

U50
ATtiny202 voltage indicator
V_IND_3V3 1 VDD
IND_LED_5V 2 PA6_LED5
IND_LED_12V 3 PA7_LED12
IND_ADC 4 PA1_AIN1
PA2_LED24 5 IND_LED_24V
PA0_UPDI 6 IND_UPDI
PA3_EXTCLK_NC 7
GND 8 GND

J1
StepStick logic row
EN 1 EN
2 MS1_NC
3 MS2_NC
4 MS3_NC
RST_NC 5
SLP_NC 6
STEP 7 STEP
DIR 8 DIR

J2
StepStick power row; VM physically omitted
GND 1 GND_VM
2 MOTOR_NC1
3 MOTOR_NC2
4 MOTOR_NC3
MOTOR_NC4 5
HOST_VIO_NC 6
GND 7 GND

J3
Keyed driver signal output
COM_PLUS 1 COM+
PUL_MINUS 2 PUL−
DIR− 3 DIR_MINUS
ENA− 4 ENA_MINUS

J4
External reference/power 5/12/24V
EXT_RAW 1 EXT+
0V 2 GND

Q1
NX6008NBK
STEP_GATE 1 G
GND 2 S
D 3 PUL_MINUS

Q2
NX6008NBK
DIR_GATE 1 G
GND 2 S
D 3 DIR_MINUS

Q3
NX6008NBK
EN_GATE 1 G
GND 2 S
D 3 ENA_MINUS

R1
1k 1%
STEP 1
2 STEP_GATE

R2
1k 1%
DIR 1
2 DIR_GATE

R3
1k 1%
EN 1
2 EN_GATE

R4
10k 1%
STEP_GATE 1
2 GND

R5
10k 1%
DIR_GATE 1
2 GND

R6
10k 1%
EN_GATE 1
2 GND

SW1
5V / EXT
VS_SELECT 1 5V
COM_PLUS 2 COM
EXT 3 V_EXT_PROT

C20
4.7uF 50V X7R 10%
BUCK_VIN 1
2 GND

C21
100nF 50V X7R 10%
BUCK_VIN 1
2 GND

C22
100nF 50V X7R 10%
BUCK_BST 1
2 BUCK_SW

C23
22uF 25V X5R 10%
VS_REG 1
2 GND

C25
4.7uF 50V X7R 10%
V_IND_IN 1
2 GND

C26
2.2uF 25V X7R 10%
V_IND_3V3 1
2 GND

D20
28V bidirectional TVS
V_EXT_FUSED 1
1
2 2 GND

D21
12V gate clamp
V_EXT_PROT 1
K
A 2 REV_GATE

D22
60V Schottky
BUCK_VIN 1
K
A 2 V_EXT_PROT

D23
60V Schottky
BUCK_SW 1
K
A 2 GND

D24
5.1V EN clamp
BUCK_EN 1
K
A 2 GND

D25
60V Schottky
VS_SELECT 1
K
A 2 VS_REG

D26
100V indicator input isolation
V_IND_IN 1
K
A 2 IND_FEED_R

F20
0.5A 63V fast fuse
EXT_RAW 1
1
2 2 V_EXT_FUSED

L20
15uH 20% Isat2.0A
BUCK_SW 1
1
2 2 VS_REG

Q20
60V reverse-polarity P-MOSFET
REV_GATE 1 G
V_EXT_PROT 2 S
D 3 V_EXT_FUSED

R20
100k 1%
REV_GATE 1
2 GND

R21
124k 0.1%
VS_REG 1
2 BUCK_FB

R22
23.7k 0.1%
BUCK_FB 1
2 GND

R23
10k 1% 0.125W
BUCK_VIN 1
2 BUCK_EN

R24
100k 1%
BUCK_EN 1
2 GND

R25
10k 1%
COM_PLUS 1
2 GND

R26
82ohm 1% 0.66W pulse-tolerant
COM_PLUS 1
2 IND_FEED_R

R27
4.7k 1%
V_IND_3V3 1
2 GND

R28
330ohm 1% 0.125W
VS_REG 1
2 GND

U20
MP2459 adjustable buck -5.06V
BUCK_BST 1 BST
BUCK_FB 3 FB
BUCK_EN 4 EN
SW 6 BUCK_SW
IN 5 BUCK_VIN
GND 2 GND

U21
AP7375 3.3V Indicator LDO
V_IND_IN 1 IN
GND 3 GND
OUT 2 V_IND_3V3

Signal connector: 1 COM+ | 2 PUL− | 3 DIR− | 4 ENA−. Common-positive drivers only; configure voltage/current and enable semantics per exact driver manual.

MCU measures COM only. 5/12/24 LEDs need firmware/calibration; neither implemented. All-off outside valid ranges. CPU <=5MHz.

UPDI TP50 | 3V3 SENSE ONLY TP51 | GND TP52. Programmer VDD disconnected; normal input supplies adapter.

Power-collapse, loaded timing, input protection, host fit and physical tests remain open. ERC is not proof of those behaviors.

Direct sinks, 5V / EXT and measured voltage indicator

PROTOTYPE — NOT RELEASED

Sheet: /

File: adapter.kicad_sch

Title: OEA external-driver adapter

Size: A2	Date: 2026-10-06	**Rev: r0.3**
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